

Mobility-aware load balancing for hybrid LiFi and WiFi networks

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Light fidelity (LiFi) is a recently proposed wireless technology that operates in a way similar to wireless fidelity (WiFi) but uses light as a signal bearer. Combining the high-speed data transmission of LiFi and the ubiquitous coverage of WiFi, hybrid LiFi and WiFi networks (HLWNets) are able to improve the system capacity of indoor wireless communications. Meanwhile, the process of access point (AP) selection becomes challenging, since the coverage areas of different networks completely overlap each other. Relying on instantaneous channel information, the conventional load balancing method assigns each user to a specific AP, providing the optimal solution for a given time instant. However, this method might cause frequent handovers for mobile users and compromise their throughputs. In this paper, mobility-aware load balancing (MALB) methods are proposed for HLWNets in both single transmission (ST) and multiple transmission (MT) modes. In the ST mode, each user is served by only one AP, chosen from either LiFi or WiFi. In this mode, the proposed method assigns each user to a certain type of network over a period of time, to suppress vertical handovers. In the MT mode, each user is simultaneously served by LiFi and WiFi, and no vertical handover occurs. A joint optimization problem is formulated to balance traffic loads between LiFi and WiFi in this mode. Results show that against the conventional load balancing method, MALB-ST and MALB-MT can improve system throughput by up to 46% and 76%, respectively, with significantly reduced computational complexity. © 2019 Optical Society of America

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1. INTRODUCTION

Global mobile data traffic is forecast to reach 48.3 exabytes per month at the end of 2021, more than four times the traffic at the end of 2017 [1]. The exponentially increasing demand for mobile data is likely to outstrip the spectrum capacity in the near future, leading to the looming crisis of a spectrum crunch in radio frequency (RF) [2]. This drives research on technologies that exploit the spectrum of higher frequencies. Among these technologies is light fidelity (LiFi) [3], which uses light as a signal bearer. Compared with wireless fidelity (WiFi), LiFi offers many advantages including (i) a much wider and license-free spectrum; (ii) secure communications, since light does not penetrate opaque structures; and (iii) feasibility in RF-restricted areas, such as hospitals and underwater. More importantly, LiFi can achieve a much higher data rate than WiFi. Recent research shows that with a single light-emitting diode (LED), LiFi is able to offer high-speed data transmission reaching 10 Gbps [4].

In comparison with WiFi, a LiFi access point (AP) has a relatively small coverage area, of approximately 2–3 m in diameter [5]. This stimulates the development of hybrid LiFi and WiFi networks (HLWNets) [6], which combine the high-speed

data transmission of LiFi and the ubiquitous coverage of WiFi. While HLWNets are able to achieve a greater throughput than stand-alone WiFi or LiFi networks [7], the process of AP selection (APS) becomes challenging. In a homogeneous network, the overlaps among the coverage areas of APs are suppressed in order to avoid inter-cell interference. A straightforward and widely used APS method is the signal strength strategy (SSS), which assigns the user to the AP that provides the highest received signal power. However, the coverage areas of different networks completely overlap each other in a HLWNet. If the SSS was used for HLWNets, WiFi would face a higher traffic load than LiFi due to a larger coverage area. Therefore, load balancing is essential for the admission control in hybrid networks. In Ref. [8], a load balancing method was proposed to achieve proportional fairness in HLWNets. Based on game theory, a distributed load balancing method was reported in Ref. [9]. A two-stage APS method was developed in Ref. [10], which first determines the users that need to be served by WiFi and then assigns the remaining users as if they were in a stand-alone LiFi network. However, the above methods all rely on instantaneous channel state information (CSI), which normally varies with time.

The indoor propagation channel varies due to two factors: (i) the motion of the portable unit and (ii) the dynamics of changing environments. The first factor imposes a much greater influence than the second one [11]. User mobility will cause handovers, which fall into two categories in a hybrid network: horizontal handover (HHO) and vertical handover (VHO). A HHO happens within the domain of a single wireless access technology, whereas a VHO occurs among different technologies. Due to the change of the air interface, a VHO requires a much longer delay than a HHO [12]. Regarding the handover mechanism for hybrid networks, the current literature is concentrated on cellular networks [13–15], where the handover process and the route-selection algorithm are the key issues. But APs are densely deployed in an indoor HLWNet, making handovers occur frequently. As a result, it is necessary to consider the handover cost when performing load balancing for HLWNets. Although the methods in Refs. [8–10] can provide an effective APS solution for a given instant, they might cause some users to repeatedly switch between different networks during the course of movement. This frequent occurrence of VHOs will compromise system throughput. In addition, fast-moving users will experience frequent HHOs in a LiFi network. To date, only a handful of studies have been conducted to address user mobility in HLWNets. Considering the handover cost, a dynamic load balancing method was proposed in Ref. [16]. This method has two limitations: (i) it is an iterative algorithm that requires quantities of iterations to reach a steady state, and (ii) it does not take VHOs into account. In Ref. [17], an efficient VHO scheme was reported, which determines whether to perform a VHO given the queue length and the condition of optical wireless channels. However, this method fails to consider the impact of HHOs on fast-moving users.

Considering both HHOs and VHOs, mobility-aware load balancing (MALB) is studied for HLWNets in this paper. Two modes are considered: single transmission (ST) and multiple transmission (MT). In the ST mode, the proposed method assigns each user to a certain type of network over a period of time, with the aim of maximizing system throughput with proportional fairness. The handover cost is considered, and it is measured through the average cell dwell time (ACDT), which is defined as the average time that a user stays connected to the same AP. In the MT mode, users are simultaneously served by LiFi and WiFi, with the facilitation of multipath transmission control protocol (MPTCP) [18]. This, however, has been widely ignored in the current literature. In this mode, the resource allocation issue is formulated as a joint optimization problem to balance the traffic loads between LiFi and WiFi. With the random waypoint (RWP) mobility model, the performance of the proposed methods is analyzed in comparison with conventional methods. Compared with the conference version [19], the new contribution of this paper is three-fold: (i) the ACDT is mathematically derived in the RWP model, (ii) a load balancing method suiting the MT mode is proposed to give more insight into MALB, and (iii) the performance of the proposed methods is comprehensively studied with different network scales. We also include results in consideration of light-path blockages. More related work can be found in Ref. [20].

Table 1. List of Acronyms

Acronym	Description	Acronym	Description
ACDT	Average cell dwell time	NLP	Nonlinear programming
AP	Access point	PD	Photodiode
APS	Access point selection	POF	Plastic optical fiber
CS	Constant speed	PPP	Poisson point process
CSI	Channel state information	PSD	Power spectral density
FLOPS	Floating point operations per second	RF	Radio frequency
HHO	Horizontal handover	RWP	Random way point
HLWNet	Hybrid LiFi and WiFi network	SINR	Signal-to-interference-plus-noise ratio
ILB	Instantaneous load balancing	SSS	Signal strength strategy
LED	Light-emitting diode	ST	Single transmission
LiFi	Light fidelity	TDMA	Time-division multiple access
LOS	Line of sight	VHO	Vertical handover
MALB	Mobility-aware load balancing	VS	Varying speed
MPTCP	Multipath transmission control protocol	VSP	Varying speed with pausing
MT	Multiple transmission	WiFi	Wireless fidelity
NLOS	Non-line of sight	WLAN	Wireless local area network

The remainder of this paper is organized as follows. The system model of HLWNets is presented in Section 2, including network deployment, channel model, user mobility model, and light-path blockage model. Related work is introduced in Section 3, and the novel methods are proposed in Section 4. In Section 5, the ACDT is mathematically derived in the RWP model. Simulation results are given in Section 6. Finally, conclusions are drawn in Section 7. The acronyms used in this paper are listed in Table 1.

2. SYSTEM MODEL

We consider an indoor HLWNet consisting of one WiFi AP and a number of LiFi APs, as shown in Fig. 1. The WiFi AP is placed in the center of the room, while ceiling LED lamps act as LiFi APs, facing downwards vertically. The WiFi AP provides coverage for the entire room, whereas each LiFi AP covers a confined area. The LiFi APs operate with a frequency reuse factor of 4, while interfering signals are treated as noise. The channel models of LiFi and WiFi are given in Sections 2.A and 2.B, respectively. At each AP, time-division multiple access (TDMA) is used to serve multiple users. The models of user mobility and light-path blockage are introduced in Sections 2.C and 2.D, respectively.

A. LiFi Channel Model

A LiFi channel consists of the line of sight (LoS) and non-line of sight (NLoS) paths. Each user is fitted with one photodiode

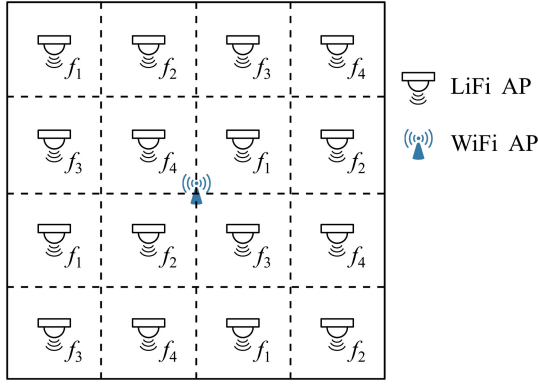


Fig. 1. System model of an indoor HLWNet.

(PD). The LoS path from AP α to user u is the straight line between them, of which the Euclidean distance is denoted by $d_{\alpha,u}$. The angles of irradiance and incidence are denoted by $\phi_{\alpha,u}$ and $\psi_{\alpha,u}$, respectively. The channel gain of the LoS path is expressed as {[Eq. (10) in Ref. [21]]

$$H_{\text{LoS}}^{\alpha,u} = \frac{(m+1)A_{\text{pd}}}{2\pi d_{\alpha,u}^2} \cos^m(\phi_{\alpha,u}) g_f g_c(\psi_{\alpha,u}) \cos(\psi_{\alpha,u}), \quad (1)$$

where $m = -\ln 2 / \ln(\cos \Phi_{1/2})$ is the Lambertian emission order, and $\Phi_{1/2}$ denotes the angle of half intensity; A_{pd} is the physical area of the PD; g_f is the optical filter gain; and $g_c(\psi_{\alpha,u})$ stands for the optical concentrator gain.

As for the NLoS paths, only first-order reflections are considered. Although higher-order reflections are important in specular cases [22], they typically contribute little in an indoor environment with few mirror-like surfaces [21]. The channel gain of the NLoS path, denoted by $H_{\text{NLoS}}^{\alpha,u}$, is given by Eq. (3) in Ref. [19].

The LiFi channel gain, denoted by $H_{\text{LiFi}}^{\alpha,u}$, can be computed as follows:

$$H_{\text{LiFi}}^{\alpha,u} = H_{\text{LoS}}^{\alpha,u} + H_{\text{NLoS}}^{\alpha,u}. \quad (2)$$

At the receiver, photons are gathered by the PD and then converted into an electric current, which is expressed as

$$I_{\text{elec}} = R_{\text{pd}} H_{\text{LiFi}}^{\alpha,u} P_{\text{opt}}, \quad (3)$$

where R_{pd} is the detector responsivity, and P_{opt} is the optical signal power. The received signal-to-interference-plus-noise ratio (SINR) of a LiFi user is written as

$$\gamma_{\alpha,u} = \frac{(R_{\text{pd}} H_{\text{LiFi}}^{\alpha,u} P_{\text{opt}})^2}{\mathcal{N}_{\text{LiFi}} B_{\text{LiFi}} + \sum_{i \in \mathcal{I}_{\alpha}, i \neq \alpha} (R_{\text{pd}} H_{\text{LiFi}}^{i,u} P_{\text{opt}})^2}, \quad (4)$$

where $\mathcal{N}_{\text{LiFi}}$ denotes the power spectral density (PSD) of noise at the receiver, B_{LiFi} is the bandwidth of the LiFi AP, and $\mathcal{I}_{\alpha}$ is the set of APs that share the same frequency as AP α . See the setup of LiFi channel parameters in Ref. [19].

B. WiFi Channel Model

The path loss model of an indoor RF propagation contains two parts: (i) the free-space loss up to a breakpoint distance, with a

slope of 2, and (ii) a slope of 3.5 after the breakpoint distance [23]. Denoting the breakpoint distance by d_{BP} , the path loss of a WiFi channel is written as {[Eq. (3.26) in Ref. [24]]

$$L(d) = \begin{cases} L_{\text{FS}}(d) + X_{\sigma}, & d \leq d_{\text{BP}} \\ L_{\text{FS}}(d) + 35 \log_{10} \left(\frac{d}{d_{\text{BP}}} \right) + X_{\sigma}, & d > d_{\text{BP}} \end{cases}, \quad (5)$$

where X_{σ} is a zero-mean Gaussian random variable with a standard deviation of σ , and the free-space loss $L_{\text{FS}}(d)$ is given by

$$L_{\text{FS}}(d) = 20 \log_{10}(d) + 20 \log_{10}(f_c) - 147.5, \quad (6)$$

where f_c is the central carrier frequency. The multipath propagation of a WiFi channel is formulated as {[Eq. (3.13) in Ref. [24]]

$$H_{\text{WiFi}} = \sqrt{\frac{K}{K+1}} e^{j\phi} + \sqrt{\frac{1}{K+1}} X_1, \quad (7)$$

where K is the Ricean K -factor ($K = 1$ when $d \leq d_{\text{BP}}$ and otherwise $K = 0$), ϕ is the angle of arrival/departure of the LoS component, and X_1 is a complex Gaussian random variable with zero mean and unit variance. The WiFi channel gain, denoted by $G_{\text{WiFi}}^{\alpha,u}$, is expressed as

$$G_{\text{WiFi}}^{\alpha,u} = |H_{\text{WiFi}}^{\alpha,u}|^2 10^{-\frac{L(d_{\alpha,u})}{10}}. \quad (8)$$

Though no interference is involved in WiFi, “SINR” is used to keep consistency. The received SINR of a WiFi user is computed as follows:

$$\gamma_{\alpha,u} = \frac{G_{\text{WiFi}}^{\alpha,u} P_{\text{WiFi}}}{\mathcal{N}_{\text{WiFi}} B_{\text{WiFi}}}, \quad (9)$$

where $\mathcal{N}_{\text{WiFi}}$ is the PSD of noise at the receiver; P_{WiFi} and B_{WiFi} are the transmit power and bandwidth of the WiFi AP, respectively. See the setup of WiFi channel parameters in Ref. [19].

C. Mobility Model

The RWP [25] is a widely used synthetic model for mobility. With the RWP, users move along a zigzag line from one waypoint to the next, with the waypoints being randomly distributed. This random motion is applied through MATLAB. Let $\mathcal{P}_{\text{WP}}$ and $\mathcal{P}_{\text{old}}$ denote the user's next waypoint and the user's current position, respectively. The movement direction is denoted by $\vec{n} = \frac{\mathcal{P}_{\text{WP}} - \mathcal{P}_{\text{old}}}{|\mathcal{P}_{\text{WP}} - \mathcal{P}_{\text{old}}|}$. The user's next position, which is denoted by $\mathcal{P}_{\text{new}}$, is calculated as follows:

$$\mathcal{P}_{\text{new}} = v \tilde{t} \vec{n} + \mathcal{P}_{\text{old}}, \quad (10)$$

where v is the user's current speed, and $\tilde{t}$ is the time resolution. When the user reaches $\mathcal{P}_{\text{WP}}$, a new waypoint will be randomly selected. Regarding the user's speed, three modes are considered.

1. Constant Speed

In this mode, each user moves with a constant speed (CS). The speeds of all users comply with a uniform distribution between 0 and v_{max} , where v_{max} denotes the maximum speed. The value

of $v_{\max}$ is assumed to be up to 5 m/s in an indoor scenario. To acquire CSI, the user's position is measured every 10 ms, during which the user can move 5 cm at most. Taking the 2–3 m coverage range of LiFi into account, this setup can guarantee a high enough resolution to track the path of movement.

2. Varying Speed

With the original RWP, users move around in a large outdoor area, e.g., a 1000 m $\times$ 1000 m region in Ref. [26], and change their speeds when they arrive at each waypoint. To render the RWP model fit for an indoor scenario, the user's speed is considered to remain the same for a short period of time. The user's movement during such a period is called an excursion. Specifically, each user moves with a randomly given speed for a random period, which is uniformly distributed between 10 s and 20 s. When the current excursion ends, the user chooses a new speed and continues moving.

3. Varying Speed with Pausing

The above two modes assume that users are always on the move. However, users might be stationary for a while. This is called pausing time, and happens when a user reaches the end of one excursion before continuing on the next one. The probability density function of pausing time is assumed to be uniformly distributed [26]. Here, the range of pausing time is set to be between 0 s and 10 s. In this mode, there are moving users and static users at the same time.

D. Light-Path Blockage

In Ref. [27], occurrence probability is used to characterize the light-path blockage at a time point. When the situation is extended to a period of time, the duration of the blockage also matters. Here, two parameters, the occurrence rate and occupation rate, are adopted to model light-path blockages. The occurrence rate, denoted by λ_u , is defined as the average number of blockages that occur in a time unit. In queueing theory [28], the Poisson point process (PPP) is usually applied to model random events such as the arrival of packages at a switch. The events of blockages are also assumed to follow a PPP. The parameter λ_u is assumed to be gamma distributed. The mean of the gamma distribution is denoted by λ . The occupation rate is defined as the proportion of time occupied by light-path blockages. This parameter is assumed to be uniformly distributed between 0 and 1. The blockages of different APs are assumed to be independent.

3. RELATED WORK

Given instantaneous CSI, the method in Ref. [8] aims to achieve a proportional fairness resource allocation. This method is referred to as instantaneous load balancing (ILB). The objective function of ILB is written as {[Eq. (14) in Ref. 8]}

$$\text{maximize } \Gamma_{\text{ILB}} = \sum_{u \in \mathbf{U}} \log(R_u^t), \quad (11)$$

where $\mathbf{U}$ is the set of users, and R_u^t is the throughput achieved by user u at the t -th instant, which can be calculated as follows {[Eq. (15) in Ref. 8]}:

$$R_u^t = \sum_{\alpha \in \mathbf{A}} x_{\alpha,u}^t r_{\alpha,u}^t p_{\alpha,u}^t, \quad (12)$$

where $\mathbf{A}$ denotes the set of APs; $x_{\alpha,u}^t \in \{0, 1\}$ and $x_{\alpha,u}^t = 1$ means that a connection exists between AP α and user u , while otherwise $x_{\alpha,u}^t = 0$; $p_{\alpha,u}^t$, a fraction variable between 0 and 1, is the proportion of time that AP α allocates to user u ; and $r_{\alpha,u}^t$ is the Shannon capacity that AP α can provide to user u :

$$r_{\alpha,u}^t = B_{\alpha} \log_2(1 + \gamma_{\alpha,u}^t), \quad (13)$$

where B_{α} is the bandwidth of AP α , and $\gamma_{\alpha,u}^t$ is the SINR of the signals that user u receives from AP α at the t -th time instant.

The ILB method assumes that the user can be served by only one AP at a time. As a result, $x_{\alpha,u}^t$ equals 1 for only one α , while others are 0. Correspondingly, $\log(\sum_{\alpha} x_{\alpha,u}^t r_{\alpha,u}^t p_{\alpha,u}^t) = \sum_{\alpha} x_{\alpha,u}^t \log(r_{\alpha,u}^t p_{\alpha,u}^t)$. Substituting Eq. (12) into Eq. (11) gives

$$\Gamma_{\text{ILB}}(\mathbf{x}, \mathbf{p}) = \sum_{u \in \mathbf{U}} \sum_{\alpha \in \mathbf{A}} x_{\alpha,u}^t \log(r_{\alpha,u}^t p_{\alpha,u}^t), \quad (14)$$

where $\mathbf{x} = \{x_{\alpha,u}^t | \alpha \in \mathbf{A}, u \in \mathbf{U}\}$, and $\mathbf{p} = \{p_{\alpha,u}^t | \alpha \in \mathbf{A}, u \in \mathbf{U}\}$.

The optimization problem of ILB is formulated as

$$\begin{aligned} & \text{maximize } \Gamma_{\text{ILB}}(\mathbf{x}, \mathbf{p}) \\ & \text{subject to } x_{\alpha,u}^t \in \{0, 1\} \quad \forall \alpha \in \mathbf{A}, \forall u \in \mathbf{U}; \\ & \quad 0 \leq p_{\alpha,u}^t \leq 1 \quad \forall \alpha \in \mathbf{A}, \forall u \in \mathbf{U}; \\ & \quad \sum_{\alpha \in \mathbf{A}} x_{\alpha,u}^t = 1 \quad \forall u \in \mathbf{U}; \\ & \quad \sum_{u \in \mathbf{U}} x_{\alpha,u}^t p_{\alpha,u}^t \leq 1 \quad \forall \alpha \in \mathbf{A}. \end{aligned} \quad (15)$$

4. PROPOSED METHODS

In this section, novel MALB methods are proposed for HLWNets in two scenarios: ST and MT. In the first scenario, the user is connected only to either one LiFi or WiFi AP. In the second scenario, the user is simultaneously served by LiFi and WiFi through MPTCP. The proposed algorithms are implemented in a central unit. Such a unit connects all APs via Ethernet cables or plastic optical fibers (POFs) [29] and acquires the required information including SINR and ACDT.

A. MALB-ST

In order to suppress frequent VHOs, MALB-ST has two stages. In the first stage, each user is assigned to a certain type of network over a period of time. Given a time instant, the second stage applies the SSS to select a specific AP within the chosen network. Denoting the type of network by κ , the handover cost of user u can be computed as follows:

$$\eta_{\kappa,u} = \begin{cases} \frac{T_{\text{HHO}}^{\kappa}}{T_{\text{ACDT}}^{\kappa,u}} & \text{if } T_{\text{HHO}}^{\kappa} \leq T_{\text{ACDT}}^{\kappa,u} \\ 1 & \text{if } T_{\text{HHO}}^{\kappa} > T_{\text{ACDT}}^{\kappa,u} \end{cases}, \quad (16)$$

where T_{HHO}^{κ} is the HHO overhead, and $T_{\text{ACDT}}^{\kappa,u}$ is the ACDT of user u when it is served by the κ -type network.

The user's average throughput, denoted by $\bar{R}_u$, can then be calculated as follows:

$$\bar{R}_u = \sum_{\kappa} x_{\kappa,u} r_{\kappa,u} \min\{p_{\kappa,u}, 1 - \eta_{\kappa,u}\}, \quad (17)$$

where $x_{\kappa,u} = 1$ means that the user is connected to the κ -type network, while $x_{\kappa,u} = 0$ means otherwise; $p_{\kappa,u}$ is the proportion of time that the κ -type network allocates to user u ; and $r_{\kappa,u}$ is the time average of $r_{\alpha,u}^t$. For an ergodic transformation, the time average of an event is equivalent to its ergodic average. Thus, $r_{\kappa,u}$ can be expressed as

$$r_{\kappa,u} = \frac{1}{N_{\text{AP}_\kappa}} \sum_{\alpha \in \mathbf{K}} \int_A B_\alpha \log_2(1 + \gamma_{\alpha,u}^A) dA, \quad (18)$$

where A is a small coverage area, $\gamma_{\alpha,u}^A$ denotes the SINR of user u at the corresponding location, and N_{AP_κ} is the number of APs in the κ -type network. Note that $r_{\kappa,u}$ depends only on the type of network, and therefore can be abbreviated to r_κ .

The term $\min\{p_{\kappa,u}, 1 - \eta_{\kappa,u}\}$ in Eq. (17) gives the smaller value between $p_{\kappa,u}$ and $1 - \eta_{\kappa,u}$. This coefficient indicates the time proportion available for data transmission. For a stationary user, we have $\eta_{\kappa,u} = 0$, and thus $\min\{p_{\kappa,u}, 1 - \eta_{\kappa,u}\} = p_{\kappa,u}$. In contrast, the value of $1 - \eta_{\kappa,u}$ becomes very small for a fast-moving user, restraining the achievable throughput in Eq. (17). Denoting the set of network types by $\mathbf{K}$, the objective function of MALB-ST is written as

$$\Gamma_{\text{MALB-ST}}(\tilde{\mathbf{x}}, \tilde{\mathbf{p}}) = \sum_{u \in \mathbf{U}} \sum_{\kappa \in \mathbf{K}} x_{\kappa,u} \log(r_\kappa \min\{p_{\kappa,u}, 1 - \eta_{\kappa,u}\}), \quad (19)$$

where $\tilde{\mathbf{x}} = \{x_{\kappa,u} | \kappa \in \mathbf{K}, u \in \mathbf{U}\}$, and $\tilde{\mathbf{p}} = \{p_{\kappa,u} | \kappa \in \mathbf{K}, u \in \mathbf{U}\}$.

The optimization problem of MALB-ST is formulated as

$$\begin{aligned} & \text{maximize} && \Gamma_{\text{MALB-ST}}(\mathbf{x}, \mathbf{p}) \\ & \text{subject to} && x_{\kappa,u} \in \{0, 1\} \quad \forall \kappa \in \mathbf{K}, \forall u \in \mathbf{U}; \\ & && 0 \leq p_{\kappa,u} \leq 1 \quad \forall \kappa \in \mathbf{K}, \forall u \in \mathbf{U}; \\ & && \sum_{\kappa \in \mathbf{K}} x_{\kappa,u} = 1 \quad \forall u \in \mathbf{U}; \\ & && \sum_{u \in \mathbf{U}} x_{\kappa,u} p_{\kappa,u} \leq N_{\text{AP}_\kappa} \quad \forall \kappa \in \mathbf{K}. \end{aligned} \quad (20)$$

The fourth constraint in the above equation indicates that for each type of network, the resource allocated to the users is no larger than the overall resource of that network. Although this is an inequality constraint, the maximization of Eq. (20) is achieved when all resources are used, i.e., $\sum_{u \in \mathbf{U}} x_{\kappa,u} p_{\kappa,u} = N_{\text{AP}_\kappa}$. Given $x_{\kappa,u}$ for each user, the Lagrange multiplier method is applicable to obtain $p_{\kappa,u}$. Let λ_κ denote the Lagrange multipliers. When $p_{\kappa,u} \leq 1 - \eta_{\kappa,u}$ for all users, the corresponding Lagrange function is written as

$$\mathcal{L} = \sum_{u \in \mathbf{U}} \log(p_{\kappa,u} r_{\kappa,u}) - \sum_{\kappa \in \mathbf{K}} \lambda_\kappa \left(\sum_{u \in \mathbf{U}} p_{\kappa,u} - N_{\text{AP}_\kappa} \right). \quad (21)$$

The solution in this case is $p_{\kappa,u} = \frac{N_{\text{AP}_\kappa}}{N_{\text{U}_\kappa}}$, where N_{U_κ} is the number of users served by the κ -type network. If any user has $\frac{N_{\text{AP}_\kappa}}{N_{\text{U}_\kappa}} > 1 - \eta_{\kappa,u}$, then $p_{\kappa,u} = 1 - \eta_{\kappa,u}$. The resource gap between $\frac{N_{\text{AP}_\kappa}}{N_{\text{U}_\kappa}}$ and $1 - \eta_{\kappa,u}$ is equally allocated to the remaining users in the same network. The optimization problem of MALB-ST can thus be solved by exhaustively searching the possible choices of $x_{\kappa,u}$. Let N_{U} denote the total number of users. The searching range of MALB-ST is only $2^{N_{\text{U}}}$, much smaller than that of ILB. Also, this method requires only one implementation as long as the ACDT does not change. The ACDT varies with the user's speed in the RWP model, which is studied in Section 5. Recalculation is needed when the ACDT changes. If the user is assigned to a different network, a VHO is then required. Otherwise, the user stays within the same network.

B. MALB-MT

With MALB-MT, each user is simultaneously served by the WiFi AP and one LiFi AP, which is determined by the SSS. Despite this, MALB-MT is a load balancing method, as it balances traffic loads between LiFi and WiFi. This process cannot be simply divided into independent resource allocation problems in LiFi and WiFi. For example, if a user has a stable and good-quality connection to LiFi as well as WiFi, it is better for the user to use mostly LiFi in order to save WiFi resources for other users. However, if the WiFi and LiFi resources are allocated independently, the resource allocation in WiFi would have no knowledge about the user's situation in LiFi.

Since the user throughput is the sum throughput provided by both LiFi and WiFi, the objective function of MALB-MT can be expressed as

$$\Gamma_{\text{MALB-MT}}(\mathbf{p}) = \sum_{u \in \mathbf{U}} \log \left(\sum_{\alpha \in \mathbf{A}_u} p_{\alpha,u}^t r_{\alpha,u}^t \right), \quad (22)$$

where $\mathbf{A}_u$ denotes the set of APs that serve user u .

The optimization problem of MALB-MT is formulated as

$$\begin{aligned} & \text{maximize} && \Gamma_{\text{MALB-MT}}(\mathbf{p}) \\ & \text{subject to} && 0 \leq p_{\alpha,u}^t \leq 1 \quad \forall \alpha \in \mathbf{A}, u \in \mathbf{U}; \\ & && \sum_{u \in \mathbf{U}} p_{\alpha,u}^t \leq 1 \quad \forall \alpha \in \mathbf{A}. \end{aligned} \quad (23)$$

Similar to ILB, MALB-MT is also based on instantaneous CSI. It is worth noting that MALB-MT does not involve the APS process, since the host APs are readily determined by the SSS. This is a nonlinear programming (NLP) problem that can be solved by the OPTI toolbox [30]. Since the user is always connected to both networks, MALB-MT can completely avoid VHOs. When a light-path blockage occurs, load balancing methods based on ST have to transfer the LiFi user to WiFi. In contrast, MALB-MT can readily shift the traffic load from LiFi to WiFi without VHOs.

5. THEORETICAL ANALYSIS OF ACDT

In this section, the user's ACDT in the RWP model is mathematically derived in order to solve MALB-ST. When the

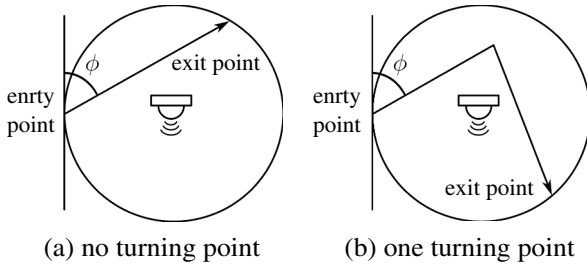


Fig. 2. Movement paths through the coverage area of an AP.

user moves through the coverage area of an AP, there are two situations: (i) no turning point and (ii) otherwise. In the first situation, the user enters the coverage area directly, as shown in Fig. 2(a). In the second situation, the user changes the direction of movement while moving inside the coverage area. Figure 2(b) exemplifies the path of the user's movement when the user changes the direction of movement once, i.e., there is only one turning point. The existence of multiple turning points is possible but less likely. If the network contains a large number of APs, the average distance of an excursion is much longer than the coverage range of an AP. As a result, users are very likely to go through several APs before reaching the next turning point. In other words, users have a much greater chance to experience the first situation than the second one. Therefore it is plausible to use the first situation to approximate the user's ACDT.

The Euclidean distance from the entry point to the exit point is denoted by l , and the angle between the direction of movement and the tangent of the circle is denoted by ϕ . With the RWP model, this angle is uniformly distributed between 0° and 90° . The average value of l is denoted by $\bar{l}$, and it can be calculated as follows:

$$\bar{l} = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} D \sin \phi d\phi, \quad (24)$$

where D is the diameter of the rounded area that an AP covers.

With straightforward integral calculations, the above equation can be rewritten as

$$\bar{l} = \frac{2}{\pi} D. \quad (25)$$

Denoting the user's speed by v , the user's ACDT is expressed as

$$T_{\text{ACDT}} = \frac{2D}{\pi v}. \quad (26)$$

In the RWP model, the user's ACDT is inversely proportional to the user's speed, which can be measured by indoor positioning systems [31]. In Fig. 3, the derived expression of ACDT is verified against simulations. When there are only a few APs, e.g., four APs, a noticeable gap exists between the theoretical analysis and the simulation, as expected from the approximation explained above. Despite this, the derived expression and the simulation have similar trends as the user's speed increases. When there are nine APs, the theoretical analysis becomes much closer to the simulation. When the number

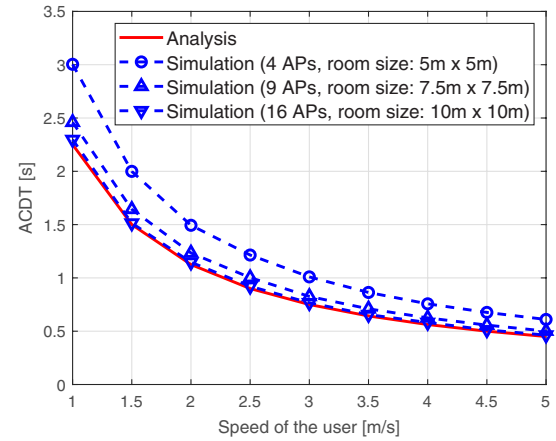


Fig. 3. ACDT of mobile users for different network scales.

of APs further increases to 16, the theoretical analysis is almost the same as the simulation.

6. SIMULATION RESULTS

In this section, Monte Carlo simulations are implemented to evaluate the performance of the proposed methods. Three baselines are considered: ILB, SSS, and MaxMin, which maximizes the minimum user throughput. The separation between the nearest LiFi APs is fixed at 2.5 m. Two network scales of LiFi, four and 16 APs, are considered, and the corresponding room sizes are 5 m × 5 m and 10 m × 10 m. Also, the vertical distance between the LiFi AP and the user is set to be 2.5 m. Considering $\Phi_{1/2} = 60^\circ$, each LiFi AP covers a rounded area with a radius of 4.3 m. This indicates a coverage overlap of 46.8% between the nearest LiFi APs. The average delay of HHO is about 200 ms in IEEE 802.11 wireless local area networks (WLANs) [32], whereas the average delay of VHO is set to be 500 ms [33]. For each case, 1000 simulations are repeated, and each simulation covers a time length of 200 s. First, we analyze the performance in terms of handover rate, throughput, and user fairness, when the CS mode is applied with no light-path blockage. Second, the performance is studied with different RWP modes. Third, a comparison of the required computational complexity is presented. Last, the effects of light-path blockages are investigated.

A. Performance in the CS Mode

The CS mode is considered with a fixed v_{max} of 5 m/s. Taking $N_U = 10$ as an example, Fig. 4 shows the rates of HHO and VHO. As expected, the proposed methods can completely avoid VHOs in the CS mode. On the contrary, ILB and MaxMin require significant numbers of VHOs. Meanwhile, MALB-MT has the largest demand for HHOs. Since MALB-ST restricts some users to WiFi only, it yields fewer HHOs than MALB-MT. When four LiFi APs are deployed, MALB-ST and MALB-MT can reduce the overall handover cost over ILB by 83% and 73%, respectively. These figures become 79% and 66% for 16 LiFi APs. This is because the proposed methods have a noticeable increase in HHOs, whereas the VHO rate increases very slightly for ILB.

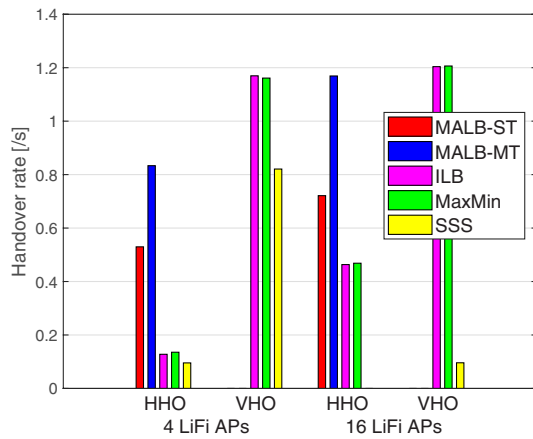


Fig. 4. Handover rate for different network scales ($N_U = 10$).

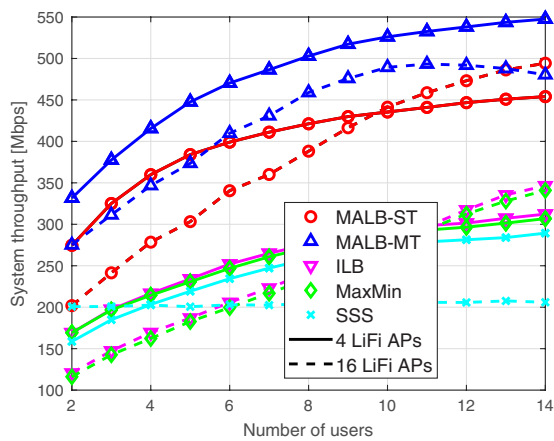


Fig. 5. System throughput versus the number of users.

The system throughput is presented as a function of N_U in Fig. 5. For four LiFi APs, MALB-MT always outperforms MALB-ST. As for 16 LiFi APs, with an increase of N_U , the throughput achieved by MALB-MT first increases and then decreases. This is because the presence of LiFi channel interference intensifies the competition for WiFi resources. Meanwhile, the throughput achieved by MALB-ST keeps increasing with N_U , since this method restricts users that have access to WiFi. In addition, MALB-ST achieves a throughput much higher than both ILB and MaxMin, which have similar performances. For four LiFi APs, MALB-ST obtains a throughput of up to 454 Mbps, 46% more than the 312 Mbps of ILB. This improvement is attributed to the reduced handover cost offered by MALB-ST. Meanwhile, MALB-MT achieves an increase of 76% in throughput against ILB, due to multiple transmissions. When the LiFi network scale increases, the performance of the HLWNet approaches that of a standalone LiFi network. Correspondingly, the gain of the proposed methods decreases. Despite this, MALB-ST and MALB-MT can still improve the system throughput against ILB by up to 42% and 38% for 16 LiFi APs.

In Fig. 6, Jain's fairness index [34] is measured for different methods. As shown, the user fairness decreases as N_U increases for ILB, MaxMin, and SSS, whereas the user fairness

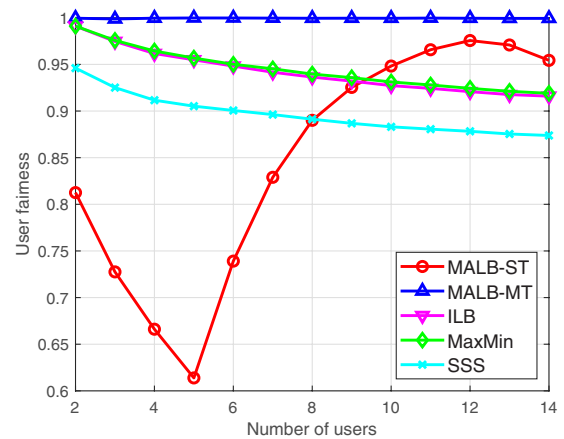


Fig. 6. User fairness versus the number of users (four LiFi APs).

of MALB-MT maintains 1. This is because MALB-MT can dynamically balance the traffic loads of the two transmission paths to provide fair throughputs for all users. MALB-ST has a lower user fairness than other methods when $N_U = 2$. As N_U increases, the user fairness of MALB-ST first decreases, then increases, and then decreases again. When N_U is larger than nine, the user fairness of MALB-ST becomes higher than that of ILB. This is because MALB-ST restricts users to LiFi or WiFi, and the users' achievable throughputs are very different when N_U is small.

B. Different RWP Modes

Figure 7 shows the handover rate in the varying speed (VS) and varying speed with pausing (VSP) modes. Compared to the CS mode, the VS mode slightly increases the VHO rate of MALB-ST, from 0 to about 0.01 per s. In contrast, MALB-MT still has no VHO, since each user is simultaneously served by LiFi and WiFi. The handover rate remains the same for ILB, MaxMin, and SSS, as their APS solutions are determined by CSI, regardless of the change in speed. When the VSP mode is applied, the handover rate drops dramatically for all methods. This is because the pausing time causes a decrease in the average speed.

In Fig. 8, the system throughput is presented as a function of $v_{\max}$ for different RWP modes. As shown, MALB-MT slightly outperforms ILB when users are stationary. When $v_{\max}$ increases, the gap between MALB-MT and ILB increases, up to 87%. Meanwhile, MALB-ST achieves a higher throughput than ILB when $v_{\max}$ is above 1 m/s in the CS and VS modes. For these modes, the average speed is equal to half of the maximum speed. Therefore the crossing point in terms of the average speed is 0.5 m/s. As for the VSP mode, MALB-ST outperforms ILB when $v_{\max}$ is above 1.5 m/s. Taking the pausing time into account, the crossing point in terms of the average speed is 0.56 m/s in this case.

C. Computational Complexity

In Ref. [8], a distributed algorithm is used to solve ILB in a complexity-efficient way, while achieving almost the same throughput. MALB-ST can also be solved in a similar way. The

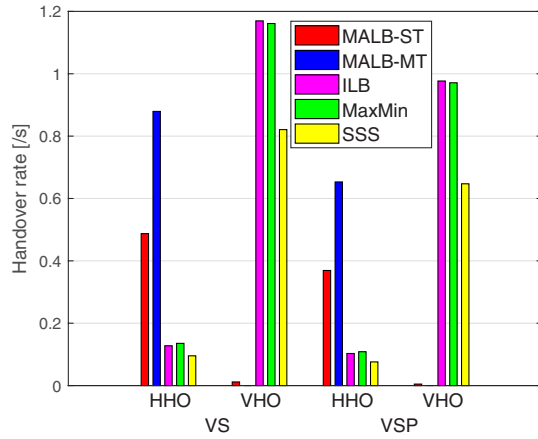


Fig. 7. Handover rate for different RWP modes (with four LiFi APs, $v_{\max} = 5$ m/s and $N_U = 10$).

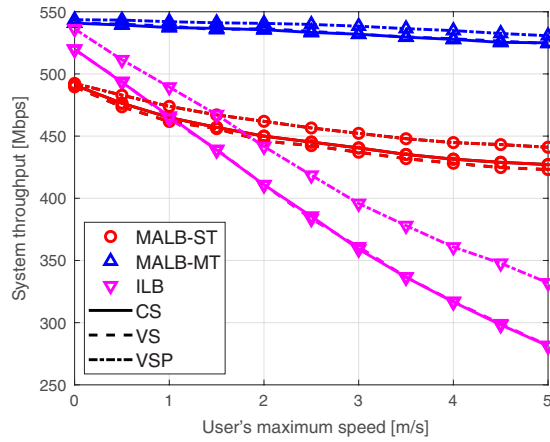


Fig. 8. System throughput versus the user's speed (with four LiFi APs and $N_U = 10$).

achievable throughput and required complexity are compared between different methods in Fig. 9. The number of users is set to be 20, in order to show the performance with respect to a relatively large N_U . Here, LiFi channel interference is not accounted for to present the interference-free system capacity. As can be seen, the proposed methods can significantly reduce computational complexity over ILB, especially for a large number of LiFi APs. This is because for MALB-ST and MALB-MT, the computation complexity is almost irrelevant to the network scale. On the contrary, the complexity required by ILB exponentially increases with the number of APs. When MALB-ST and ILB are both solved by the distributed algorithm, their computational complexities are close. However, MALB-ST can greatly improve the system throughput against ILB.

D. Impact of Light-Path Blockages

As shown in Fig. 10, in general, the system throughput decreases as the occurrence rate of light-path blockages λ increases. The only exception is that when $v_{\max} = 5$ m/s, the throughput achieved by ILB first increases and then decreases. This is because ILB relies on instantaneous CSI, and its

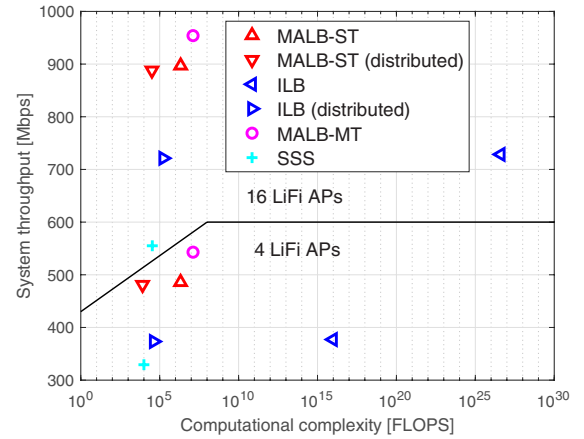


Fig. 9. System throughput versus computational complexity (in the VSP mode with $v_{\max} = 5$ m/s and $N_U = 20$).

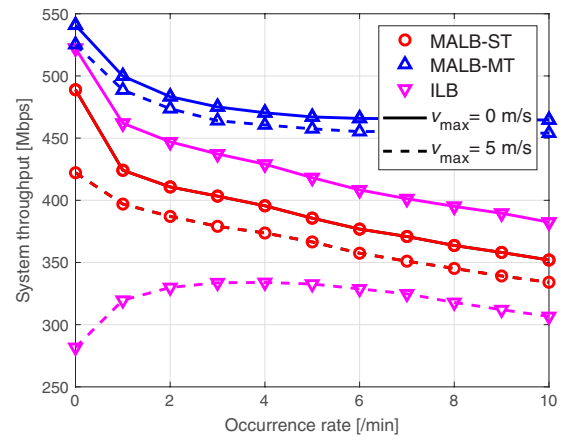


Fig. 10. System throughput versus the occurrence rate of blockages (in the CS mode with four LiFi APs and $N_U = 10$).

handover rate decreases due to occasional light-path blockages. This mitigates the throughput degradation caused by handovers. It is also found that the throughput of MALB-MT decreases to saturation as λ increases. This is because using this method, all users have access to WiFi without handover when light-path blockages occur. Therefore, the performance of MALB-MT is limited by the occupation rate instead of the occurrence rate.

7. CONCLUSION

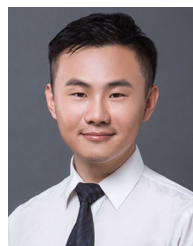
In this paper, MALB was investigated for HLWNets. Two novel methods, MALB-ST and MALB-MT, were proposed for the ST and MT modes, respectively. Based on ACDT, MALB-ST assigns each user to a certain type of network over a period of time. During this period, the SSS is applied to select specific APs, and users can transfer between APs in the same network. MALB-ST can suppress VHOs caused by user mobility, at the cost of an increased HHO rate. In the MT mode, users are simultaneously served by LiFi and WiFi, and VHOs can be completely avoided. MALB-MT performs a joint resource allocation to balance the traffic loads between LiFi and WiFi.

Results show that MALB-ST achieves a higher throughput than ILB when the user's average speed is above 0.5 m/s. When the user's average speed reaches 2.5 m/s, MALB-ST can improve the system throughput over ILB by up to 46%. MALB-MT works much better than MALB-ST when there is no interference among LiFi APs. Compared to MALB-ST, MALB-MT achieves a larger gain over ILB, of up to 76%. Future research will involve research into practical behaviors of user movement as well as the resulting changes in performance.

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